BLACK PEN Answer all questions first. **GREEN PEN** Then: tick and rewrite mistakes.

LESSON 144 · Lesson 144 · DNS and IP Routing: From Name to Destination

Name: _____ Class: _____ Date: _____

Work through the pages in order. Answer in BLACK PEN first. Check in GREEN PEN only after the Exit Ticket.

Do Now · Recall previous learning

Q1. Which TCP/IP layer handles routing and IP addressing?

Q2. What data structure uses LIFO?

Q3. What is a trace table used for?

Q4. How many bits are in one byte?

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick and rewrite mistakes.

I Do · Read new knowledge

Key vocabulary

DNS — Hierarchical name-resolution system.

IP address — Network-layer address used to route packets.

resolver — Client/service that queries DNS and caches answers.

router — Forwards IP packets towards destination networks.

New learning

Resolve hostname — Browser requests address for a website name through DNS.

Use IP destination — Application connects to a resolved IP address and transport port.

Route across networks — Routers forward packets hop by hop to the destination network.

Worked example

A pupil types example.org and receives an IP address from DNS. Explain what DNS does and does not do.

1. Name example.org — Identify the relevant information and make a prediction.

2. DNS returns an IP address — Explain the change using correct technical terms.

3. Connect to web server — DNS maps the hostname to relevant resource records such as an A/AAAA address. DNS does not directly transmit the web page. Browser makes a separate connection to the chosen server address and requests content, often over HTTPS.

We Do · Practise with guidance

A DNS answer has been cached but website server has moved. Explain a possible issue and resolution.

Cached old address

Identify the relevant information and make a prediction.

Server address changes

Explain the change using correct technical terms.

Refresh after TTL

The cached address may remain until its TTL expires; client may contact an outdated endpoint. The resolver can refresh after expiry. DNS propagation and authoritative changes do not guarantee immediate worldwide consistency.

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick and rewrite mistakes.

You Do · Work independently

BLACK PEN: Write your own answers. Do not look at the Review & Improve pages yet.

Question 1

Explain how DNS helps a user retrieve a website by name. [6]

Question 2

Differentiate DNS name resolution from IP packet routing. [6]

Question 3

A school website moves servers but some users reach the old address. Explain cached DNS and propose sensible checks. [8]

Exit Ticket · Answer before checking

DNS sends the entire web page to the browser after it receives the hostname.

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick and rewrite mistakes.

Review & Improve · GREEN PEN

Now change to GREEN PEN. Find each original question. Tick correct answers; read the model and rewrite mistakes.

Page 1 onwards · Do Now

Q1. Internet layer.

Q2. Stack.

Q3. Tracking variable state changes as an algorithm runs.

Q4. Eight.

You Do pages · Independent questions

Q1. Hostname lookup [2], address records returned/cached [2], browser uses IP to connect to server [2].

Q2. DNS maps names/records [3], routers forward packets using IP destination/network information [3].

Q3. Resolver caching [2], TTL expiry [2], changed authoritative record [2], client/network diagnostic tests [2].

Exit Ticket · Model answer

False. DNS supplies name-resolution data; web content is retrieved separately.

Try the question again

Choose one question you got wrong. All correct? Choose the hardest.

1. Copy the whole original question:

2. Write a complete new answer in GREEN PEN. Show working if needed: